

Annex B

Cover Page for JJC 2009 Maths H2 Prelim Exam Papers

Qn/No	Topic Set	Paper N0	Answers	Marks
1	Curve Sketching 1 – Transformations	9740/01	-	6
2	Functions	9740/01	(i) $R_f = [2, 4)$ (ii) $gf : x \mapsto 1 + \frac{1}{3 - 2e^x}, \quad x \in \mathbb{R}, x \leq 0$ (iii) $R_{gf} = \left(\frac{4}{3}, 2\right]$	7
3	Complex Numbers	9740/01	$w = 2\sqrt{3} + 4i$	7
4	Differential Equations	9740/01	(i) $P = 100(270e^{0.01t} - 260)$ (ii) $t > 3.64 \text{ years}$	7
5	Integration - Techniques	9740/01	(a) $\frac{1}{2}[\tan^{-1} 2x] - \frac{1}{2}\ln(4x^2 + 1) + c$ (b) $\frac{1}{2}\left[\frac{\pi}{4} + \frac{1}{2}\right]$	8
6	(a) System of Linear Equations (b) Inequalities	9740/01	(a) 75 chicken, 125 horses, 150 sheep. (b) $x < -3$ or $0 < x < 5$. $0 < x < e^{-3}$ or $1 < x < e^5$.	10
7	(a) Mathematical Induction (b) Method of Difference	9740/01	(b) $1 - \frac{1}{(N+1)!}$	10
8	AP, GP & Sigma Notation	9740/01	(i) $r = \frac{1}{3}$ (ii) least $N = 12$	11
9	Vectors	9740/01	(ii) $\begin{pmatrix} 4 \\ 5/2 \\ -1/2 \end{pmatrix}, \frac{\sqrt{14}}{2}$. (iii) $r = \begin{pmatrix} 8 \\ 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix}, \text{ where } \lambda \in \mathbb{R}.$	11
10	Complex Numbers	9740/01	(a) $w = -1 - i\sqrt{3}, z = \sqrt{3} + i$. (b) $2e^{i\left(-\frac{5\pi}{8}\right)}, 2e^{i\left(-\frac{\pi}{8}\right)}, 2e^{i\left(\frac{3\pi}{8}\right)}, 2e^{i\left(\frac{7\pi}{8}\right)}$. (c) $2\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right),$ $\{n : n = 3(2c - 1), c \in \mathbb{R}^+\}$	11
11	Differentiation & Applications	9740/01	(a) $t = -\frac{1}{8}$. (b) $r = \frac{h}{2(h-1)}.$	12

	Pure Maths			
1	Vectors	9740/02	(ii) $\begin{pmatrix} 15 \\ 11 \\ 4 \end{pmatrix}$. (iii) $\frac{\sqrt{371}}{30}$, $3\sqrt{371}$.	8
2	Integration - Applications	9740/02	(a) $\frac{1}{n^3} \sum_{r=1}^{n-1} \left(r^2 e^{\frac{r}{n}} \right)$, $e - 2$. (b) 19.77	8
3	(a) Binomial Series (b) Maclaurin's Series	9740/02	(a) $\frac{1}{2^{n+3}} (8 - 28nx + 49n^2 x^2 - 35nx^2)$, $-2 < x < 2$. (b) (ii) $1 + \frac{x}{2} + \frac{x^2}{8} - \frac{7x^3}{48}$. (iii) $1 + \frac{x}{2} - \frac{3x^2}{8}$.	11
4	Curve Sketching 2 – Rational Functions	9740/02	(iv) 2	13
	Statistics			
5	Sampling	9740/02	(i) non-random,/biased. (ii) Stratified sampling.	4
6	Hypothesis Testing	9740/02	(i) p-value = 0.0427, Reject Ho.	7
7	Permutations & Combinations	9740/02	(a) 840. (i) 30240. (ii) 302400. (b) 66	7
8	Probability	9740/02	(a) (i) A and B are independent. (ii) 1/3. (iii) 3/4. (b) $\frac{3}{595}$. (c) $\frac{52}{1683}$.	10
9	Binomial & Poisson Distributions	9740/02	(a) 0.244. (b) 149. (c) 0.910.	10
10	Normal Distribution	9740/02	(a) 8.1 (b) 3.4 (c) 0.955 (d) 0.844	11
11	Correlation & Regression	9740/02	(i) $y = -\frac{1}{c}x + \frac{\ln k}{c}$. (ii) $r = -0.994$ (iii) $y = 2.14 - 0.662x$, $V = 2.2$, (iv) $c = 1.51$, $k = 25.5$. (v) 0.0282.	11